Tracking Events Through a News Stream: A Filter, and the Association Layer It Needed

Working paper. Rewritten for length 2026-08-25 — the previous 1,033-line chronological version is at git `1e27878`*. Every measured result and every negative below is carried over; what was cut is narration, not findings. The build record with contemporaneous detail is* `EKF_MHT_BUILD_RECORD.md`.

Abstract

A real-world event reported across a stream of news documents has a state that evolves: a death toll rises, gets revised downward, and is reported by sources that disagree, censor ("at least 12"), and lag. We build the pipeline — synthetic stream generator, LLM realizer, schema-driven extraction, normalization, and a per-role Extended Kalman Filter — and the programme's question has two halves: **track** an evolving quantity, and **diarize** observations into the right event stream.

The filter half was solved early and validated on held-out synthetic data, where fine-tuning the extractor closes **75% of the gap to the structured-observation ceiling**. The diarize half then took eleven weeks, produced more negative results than positive ones, and is the subject of this paper.

It is now solved, and not by the mechanism the design specified. A global Viterbi decode over three states — *own place, aggregate, reject* — improves every event we have at one setting:

Pooled RMSE, in deaths — lower is better.

event	ratio gate	global decode	change	three-way oracle
Hurricane Helene 2024	29.3	**20.7**	−29.4%	17.6
Türkiye–Syria 2023	11,581.5	**10,695.5**	−7.6%	3,287.4
Aegean Sea 2020	74.4	**15.7**	−78.8%	19.1

(σ=0.3, reject_cost=4.0, stay=0.1, warmup=0. The oracle column consults ground truth and cannot ship; it prices the ceiling. On Aegean the decode beats it because that oracle applies a fixed drop rule rather than optimising the estimate — see §7.)

Metrics used in this paper

Two scales appear, and they are not comparable. **Anything quoted below states which.**

metric	units	meaning
pooled RMSE	deaths	one RMSE over every (stream, time-grid) point. **The headline.**
nRMSE	dimensionless	per-stream RMSE normalised by that stream's range, then macro-averaged
binding precision	%	of the figures a model binds to a role, the share that are the *correct* figure
strict F1	—	exact span *and* role match, no partial credit
catch / false-reject	counts, %	cross-event figures rejected, against genuine observations wrongly rejected

nRMSE was the headline until the metric error in §6: normalising per stream lets a stream with a tiny range outvote the largest one, which is how a correct mechanism was recorded as a failure. Pooled RMSE in deaths replaced it. **Numbers from before that correction are in nRMSE and are marked as such** — they are kept because the arguments built on them are still the arguments, and re-running every historical arm on the new metric was not affordable.

Two properties carry it, and the specified design — a hypothesis tree with Hungarian assignment — is neither. The decision must be **global**, because a greedy rule commits per observation and one large figure admitted early poisons a stream's running scale for everything after it. And it must be able to **reject**, because assignment headroom is measured at **zero**: a two-way oracle scores exactly what the shipped gate does, and the entire residual is the null hypothesis. Hungarian assignment optimises the half worth nothing.

The paper's second contribution is the negative space around that result. Four alternatives were built and measured against the same events and all lost; a magnitude gate that won 9× on the event it was designed against split held-out; a corpus built to teach cross-event suppression was beaten by a one-line threshold; and four separate instrument defects produced confident wrong answers that survived because the instrument returned the expected result. §6 is that list, and it is the more transferable half.

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1. The problem

Two halves, co-equal, and the second is not downstream of the first:

half	mechanism	status
track — recover an evolving quantity from noisy, censored, lagged reports	Extended Kalman Filter	built, validated on synthetic
diarize — decide *which* event stream each observation belongs to	global decode over	**built 2026-08-25**; §4

"Diarize" is borrowed from speaker diarization. It is not "who spoke when" but *which event is this figure about, over time*, and naming it that way makes obvious that the two halves fail independently and that the second can silently destroy the first.

2. Method

Per role (`dead` , `injured` , `missing`), a 1-D random-walk Kalman filter with **relative** measurement noise: `R = (σ · max(ref,1))²` where σ comes from the source (official 0.06 / major outlet 0.12 / preliminary 0.25) scaled by the qualifier (point 1.0 … feared 2.5). Process noise grows as `q_rel · max(μ,1) · dt`, so genuine jumps stay admissible between reports. Noise scales by the current **estimate**, not the observed value — scaling by the raw report over-trusts a low reading and drags a rising estimate down.

Observations come from a four-stage pipeline: a relevance gate, event extraction, casualty record extraction, and normalization to `(value, qualifier, source)`. `GATES.md` documents every filter, its type, and its default.

3. What was established before the association work

Extraction is solvable. On held-out synthetic streams, fine-tuning the extractor closes 75% of the gap between zero-shot extraction and a structured-observation ceiling. The generator, realizer and blind-test protocol are in the build record.

On real news the tracker lost to a trivial baseline. On the pre-registered Türkiye–Syria 2023 run, `est_last_value` beat the EKF (nRMSE 0.208 vs 0.136), a 1999 death toll quoted in an article's history section was tracked as a 2023 figure, and one of the two affected countries was never recovered at all. **Attribution, not filtering and not extraction, was the bottleneck** — and that diagnosis has held ever since.

A magnitude scope gate won 9× and split held-out. Every state stream in the Helene run is contaminated, and always upward: North Carolina (true peak ~123) receives 200, 215, 227, 230 and 250. Walking a stream in time order, the tracker's own running estimate supplies a scale, and a reading far above it is not a rival claim about the state but an observation about a larger scope. Reclassify rather than discard — a rejected figure moves to `__aggregate__`, where it is a correct observation instead of a corrupting one. Pooled error on Helene: **314.5 → 29.3 deaths**.

It then split on held-out data: 3.7× better on the contaminated stream, 2.3× worse on the clean one (nRMSE, per stream). The reason is diagnostic and set up everything after it — **without a declared scope hierarchy, "a larger scope" and "the largest part" are indistinguishable from the numbers alone.** We declared one (`rollup.json`), and the reference it enables turned out to be a measured negative in its own right (§6).

4. The association layer

4.1 The oracle that priced it — and priced it wrong first

Before spending on a hypothesis tree, we priced it. Assign every observation to the scope it actually fits, using ground truth: a ceiling, not a method.

```
shipped scope gate      0.591   nRMSE
two-way oracle          0.537   nRMSE
headroom               +0.055   (9.3% relative)
```

That number prices the wrong thing, and the tell was already in the table. The oracle is *two-way* — every observation goes to its own place or to the national total — so a figure belonging to no scope in this event has no correct home, and it scores Katrina's 1,400 exactly as badly as the shipped gate does. On two states the gate already **beat** the "perfect" assignment, because it has a third option the oracle lacks: *drop*. That was read as a curiosity rather than as a defect in the instrument.

Give the oracle a reject option and the ceiling moves to **+0.111 (18.8%)** — double. Later, re-derived on the pooled metric: shipped gate 29.3 deaths, three-way oracle 17.6, and it gets there **purely by dropping more** (106 observations kept down to 76).

So assignment headroom is zero and the whole residual is the null hypothesis. Both properties of the eventual fix are in that sentence, and it took two more failed builds to read them.

4.2 What the reject option is actually rejecting

Restricting to the gated population — the two-way oracle keeps everything else by construction, which was inflating a `gated` feature to AUC 0.763 for free — the oracle's rejects have **median value 6.0** against the kept set's **99.0**. Split by direction against the place's own truth:

	share
stale, BELOW truth	**63%**
over, above truth	20%
no truth coverage	17%

The gate rejects *upward only*. It is blind to two thirds of its own prize by construction — an article at t=83.8h still saying "three" dead in North Carolina when the toll is 46.

4.3 Track birth, built and lost — and what the failure localized

The cheapest piece of MHT that delivers a null hypothesis is track birth by innovation gating. It was built and **lost to the fixed magnitude ratio it was meant to replace** (nRMSE 0.608 against 0.591), degrading the national stream 6.7× while doing so, because judging a stream against its own track is circular in exactly the way the scope reference was.

The missing property was not a better birth rule. It is that **the decision has to be made over the whole sequence at once.**

4.4 The global decode — what ships

`scope_gate.hmm_gate`. Three states, decided by Viterbi over the stream, with hard-EM around the decode because the `own` level is itself unknown.

Emissions are **one-sided**: a rising toll may legitimately exceed the level established so far, so only a reading far *below* it argues against `own`. The band top is `natl/part_ratio` — the same boundary the ratio gate uses — so this is a soft form of §3's rule rather than a rival to it. Per-observation evidence from outside the magnitude channel (an out-of-window date, a place outside the declared hierarchy, a syndication marker) is added to the reject state, so those filters **argue rather than veto**.

Results are in the abstract. On Helene it beats the gate at all twelve knob settings swept and captures 73% of the measured reject headroom; on Aegean the ratio gate is *inert* — 74.4 at ratio off, 4.0, 3.0, 2.0 and 1.5 alike — so that column is not a tuning margin.

Design rule, measured: keep every feature weight **below `reject_cost`**, so no single feature can force a reject alone — it can only tip a case magnitude has already made marginal. The sweep shows a cliff exactly at that boundary, and it is what protects Türkiye.

It nearly read as another split result. The first sweep had Türkiye worse, which would have been the third intervention in a row helping Helene and hurting Türkiye, and we proposed the divergence was structural — Türkiye's `global-max` reference is self-referential for its dominant stream. **The per-stream diagnostic refuted that**, in the opposite direction: the decode helped the self-referenced dominant stream (nRMSE 0.403 → 0.377) and hurt the *minor* one (0.923 → 1.858). Chasing the inversion found the cause — a `warmup` parameter copied from the greedy gate, which pins the first readings to `own`. Syria's first two are contaminating Türkiye figures (9,057 and 17,674 against a true peak of 5,800); pinning them poisoned the level and the genuine 3,317s were then rejected as stale. **That is precisely the greedy commitment the global decode exists to remove, smuggled back in as a knob.**

4.5 Four alternatives, all measured, all rejected

- **Student-t measurement model.** One-sided IRLS reweighting. Helene −1.7 deaths, Türkiye +651. Retires none of `REJECT_SIGMA`, `MAX_RATE` or the gate's drop branch, which was the reason to want it: with the gate off, Helene is 314.5 deaths under every ν tested. **Whatever the gate catches, it is not a fat tail.** The symmetric textbook form is much worse on both events, so the one-sidedness is measured rather than assumed.
- **Soft PDA association.** Loses to the hard decode on both events (pooled RMSE, deaths: Helene 28.7 vs 20.7; Türkiye 19,150 vs 10,696). Two structural reasons: soft weighting cannot **remove**, and the headroom is in removal; and PDA arbitrates *independent* targets while ours are **nested** — a place's toll is part of the national toll — so an ambiguous reading is soft-assigned to both part and whole at once (106 observations became 118 assignments).
- **A fourth state for downward reclassification.** Correct and **inert**. Helene's North Carolina truth falls four times, but the reports never follow it down: after each revision the later readings are 230, 230, 230, 1,400, 98, 250 while the official toll falls to 84–123. No state, filter or change-point detector can track a revision that is never reported. This reframes `CENSOR_AT_LEAST` — the filter is not wrong to refuse to descend, **the data never descends** — making it a source-acquisition problem, not a modelling one.
- **Folding the date and scope gates into the emission.** Works on its own terms — on Helene, cross-event rejection goes from 5/6 caught at 19.8% false rejections to 4/6 at 9.9%, at no trajectory cost —

and cannot be validated on the headline metric. See §7.

5. The front end, and why it moved onto the critical path

The residual under the scope gate is **4.7% cross-event contamination**, and no decode-side signal worked against it. The corpora offered no purchase either: **0.0% of training documents contain a figure the model is supposed to leave alone.**

So we built that corpus — withholding an interfering event's records while keeping its text — and trained on it. The suppression is real: it removes 15 of 20 large false positives and more than halves the ungated error. **Then a declared per-event plausibility ceiling — one threshold, no model, no training — beat it** (ungated per-place mean nRMSE 378.809 → 18.287 on the production model), because the large false positives were never other storms' tolls. They were Asheville's population, FEMA flood-insurance policies, power crews, and years read as death tolls. Both genuine cross-event figures survive muting *and* the ceiling.

The remaining candidate was a span-embedding router, and it was **blocked on the front end**: no model then available emitted trigger-and-argument spans on wire copy at all. That put the extractor on the critical path, and it was rebuilt (mmBERT, English trigger→argument 798 → ~39,800 examples). The rebuild **works**: it binds the correct death toll 67–100% of the time (binding precision) against the incumbent's 0–7.7%, and beats it on all eight strict F1 heads of the shared blind test.

Its pre-registered gates 1 and 2 failed anyway, and gate 1 was the instrument. Gate 1 counts a window as satisfied when a trigger and *any* bound argument appear, never checking whether the bound figure is right — the incumbent's "65%" was 39 firings carrying **three** correct tolls. Scored best-over-a-range, a form gate rewards indiscriminate firing. Gate 2 — that two same-type events in one passage stay separable — fails for both models and is a property of the decode, not the corpus: the record head pools same-type instances into one. That remains open.

6. Negatives, and what each one closed

- **Our strongest prior claim does not reproduce.** Recorded in the build record; it did not survive re-measurement on a clean run.
- **Aggregates constrain the sum, and our metric only scored the split.** §6.2's original verdict — an aggregate row "does not pay off" — was decided by a scorer that measures only the split. Re-scored in absolute deaths, **the aggregate improves the national total at every density** — 87.6 → 28.5 at 10% part density — most where the paper said it "loses worst". The mechanism was right and the verdict was wrong, from the same sentence. The conclusion had survived three deliberate robustness attacks in two days — proportional Q, correlated Q, and a seed/noise floor — none of which could find the problem, **because all three varied the model and none varied the scorer.**
- **Restructuring the text does not fix attachment.** Rewriting articles into self-contained bullets does not repair number-to-place binding.
- **Guide-filtered negative sampling: wired, tested, negative.** The guide veto lost on 7 of 8 metrics.

- **The implied-max reference loses badly** (nRMSE 2.590 vs 0.591 on Helene) — the same circular self-reference as track birth.
- **Declared knowledge beats learned signal at cross-event rejection.** Scope membership, with zero model calls, catches 4 of 6 cross-event figures at **7.3% false positives**; the best learned signal tried, one call per observation, catches the same 4 at **31.7%**. Same recall, a quarter of the false positives, no model.

7. Limitations

- **The gate collapse cannot be validated on the headline metric, on any feed we have.** Folding the date gate, the scope gate and syndication boilerplate into the emission works on its own terms (Helene: 5/6 caught at 19.8% false → 4/6 at 9.9%, no trajectory cost) but moves pooled RMSE by **nothing** — on Helene, and on the Aegean event built specifically to test it (15.74 at every feature weight from 0 to 3, while demonstrably rejecting three of six contaminants). The reason is structural: **association routes contaminants out of the scored streams before the gate ever sees them.** Aegean's 17,000s are keyed `unknown` and `marmara`, never `Izmir`, and only 12 of 53 observations are bound to a gated place at all. A fourth event would not change this.
- **A pre-registered prediction failed, and the failure was informative.** Aegean was chosen for scale separation — 119 deaths against an İzmit 1999 reference of ~17,000, 143× where Helene's contaminants differ by 6× and Türkiye's by 3×-and-crossing — with a registered prediction that the collapse would show a large gain. It did not, for the reason above, which has nothing to do with scale separation.
- **Two same-type events in one passage remain inseparable.** A property of the record head, which pools them into one instance. No corpus fixes it; the instance dimension is the next constraint.
- **Four instrument defects produced confident wrong answers.** A threshold sweep that was inert for five measurements; a strict-vs-relaxed comparison that turned a doubling into an apparent halving; gate 1 counting firings as though they were hits; and a metric that scored only the split. **Each survived because the instrument returned the expected answer** — a harness that "correctly" fails the incumbent is not audited. Pre-registration constrains the threshold, not the measurement apparatus.
- **The exposure/casualty boundary is unmodelled.** 11% of Helene's `dead` observations are hand-audited non-casualty; half are exposure counts (300 rescued, 50 patients rescued, 32 evacuated) and half unit confusion (a two-day period, six states, 1,400 landslides). A rescued person is a *counterfactual* casualty. The schema has four roles — location, injured, missing, dead — and none for exposure, so a number with no correct home lands in a wrong one. That is a data-modelling gap, not a gate.
- **The relevance gate does not filter non-English at all.** The shipped model admits 199 of 200 clean Turkish articles, because it is English-only by vocabulary — worse than the 58.5% that forced an earlier rewrite, and silent. Translation does not fix it.
- **Conflict casualties are out of scope for validation.** The taxonomy extends (KIA/WIA are realized harm; belligerent side is the scope axis again), but conflict ground truth is contested by construction

and could not be scored the way these three events are.

8. Reproducibility

The Helene observation set cannot be regenerated. It depends on archived captures whose availability has since changed; the extracted observations and the ground-truth trajectory are committed, the article text is not (it belongs to its publishers) and lives in a gitignored cache. The Türkiye feed reads truth from the same sentence the extractor reads, which is why `est_last_value` scores nRMSE 0.000 there by construction — a defect the Helene and Aegean feeds were built to avoid. **Aegean is the only feed with genuinely independent sources on both sides**: ground truth from the Wikipedia revision history (55 timestamped points, İzmir 12 → 116, including a real downward reclassification), documents from Turkish English-language wire copy.

Code, feeds and every sweep referenced above are in `tools/ekf_showcase/`, indexed in its README with the verdict attached to each probe. Gate defaults and types are in `GATES.md`. Contemporaneous build detail is in `EKF_MHT_BUILD_RECORD.md`; the running narrative, including the errors as they were made and caught, is in `PROJECT_JOURNAL.md`.

9. References

Carion et al., *End-to-End Object Detection with Transformers* (2020) — the set-prediction framing and the Hungarian matching argument that prompted the global-decode design, and whose §4.2 "why greedy fails" is the same argument as §4.3 here.